appear. However, if we instead set $\vartheta = \vartheta_2(\boldsymbol{q})$, then

$$\begin{aligned}\langle \vartheta_2(\boldsymbol{q}) f_1(\boldsymbol{q},\boldsymbol{p})\rangle &= \pm\frac{\beta}{2}\left\langle (\gamma U(\boldsymbol{q}) - c)\vartheta_2(\boldsymbol{q}) \pm \boldsymbol{p}^T \boldsymbol{M}^{-1}\nabla U(\boldsymbol{q})\vartheta_2(\boldsymbol{q})\right\rangle \\ &= \pm\frac{\gamma\beta}{2}\left\langle (U(\boldsymbol{q}) - \langle U(\boldsymbol{q})\rangle)\vartheta_2(\boldsymbol{q})\right\rangle,\end{aligned}$$

in which case our correction becomes $\mathcal{O}(\gamma)$. This is potentially disastrous if we choose the friction to be large (consider $\gamma \sim h^{-1}$), as the error introduced through discretization will dominate the true average to be sampled. Examining the algorithms themselves this is unsurprising, as although we have integrated the Ornstein-Uhlenbeck process exactly, when considering the limit of infinite friction in the ⟦XYZ⟧ schemes we obtain an inconsistent discretization. Hence these first-order schemes will be unsuitable for the integration of overdamped Langevin dynamics.

This is not always the case for first order schemes, however. For example the scheme denoted ⟦AOAB⟧ is consistent with the dynamics, and may be used for integrating the overdamped ($\gamma = \infty$) Langevin equation.

7.8.2 Numerical Experiment

Let us consider testing the computed first order corrections on a one dimensional example. Let $M = \beta = 1$ and define the biased double-well potential as

$$U(q) = (q^2 - 1)^2 + q/2.$$

We shall consider computing the average of a specific observable $\vartheta(q,p)$, and estimating the error due to discretization using the correction functions given in (7.25), where we choose the observable

$$\vartheta(q,p) = p^2 - qU'(q) + 2qp, \qquad \langle\vartheta\rangle = 0.$$

The low dimensionality of the problem and the simplicity of the observable and potential energy function make resolving the perfect sampling bias to leading order an easy task, allowing us to give a good estimate (to order h^2) of the discretization error. In Fig. 7.4 we plot the results of $\langle\vartheta\rangle_h$ computed via numerical experiment, using each of the six first order schemes at either $\gamma = 1$ or $\gamma = 2$, and compare them to our first order prediction

$$\langle\vartheta\rangle_h \approx \langle\vartheta\rangle + h\langle f_1\vartheta\rangle.$$

The numerical experiment at each timestep is averaged from 16 independent experiments, each running over a time interval of 10^8. The results match our